

PINN v4

Physics-Informed Neural Network for
Non-Invasive Diastolic Hemodynamic Estimation

Kwanhyeong Lee

Architecture & Design Features

The Problem: Nagueh Tautology

v3c — Circular Dependency

Simulator:

EDP (set)

$$\rightarrow E/e' = (EDP - 1.9) / 1.24$$

PINN learned:

E/e' (input)

$$\rightarrow EDP = 1.24 * E/e' + 1.9$$

Result:

PINN merely reproduced the Nagueh formula.

EDP AUC = E/e' AUC $\rightarrow$ zero added value



v4 — Physics-Based Solution

E/e' is EMERGENT from physiology:

$$e' = f(\tau, E_{ed})$$

relaxation + stiffness

$$E = g(\Delta P, \tau)$$

transmitral gradient

$$E/e' = E / e' \text{ (not prescribed)}$$

Same $E/e' = 10$ can mean:

$\tau=45, E_{ed}=0.06 \rightarrow EDP=10$ (normal)

$\tau=55, E_{ed}=0.04 \rightarrow EDP=22$ (HFrEF)

Architecture Overview

11 Inputs

EF
EDV, ESV
SBP, DBP
HR, CO
 E/e'
E/A, DT
LAVI

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Encoder MLP

Linear(11, 256)
SiLU + BN + Drop

Linear(256, 128)
SiLU + BN + Drop

Linear(128, 64)
SiLU + BN

Linear(64, 7)

>>

7 Latent

Ees_delta
 E_{ed}
V0
A
 τ
Ea_raw
gate_Ea

>>

Physics Decoder

Ees
 E_{ed} (stiffness)
Ea (afterload)
Ea/Ees coupling
EDP (mmHg)
Efficiency
 τ (ms)

Raw inputs also feed decoder directly (bypass connection for physics equations)

Physics Decoder: EDPVR Embedding

$$EDP = A \times \exp(E_{ed} \times (EDV - V_0))$$

Pure exponential EDPVR — no Nagueh formula blending

E_{ed}

[0.005, 0.50] 1/ml

Chamber stiffness
coefficient.
Key diastolic
biomarker.

V₀

[-20, 30] ml

Unstressed volume.
Derived from
ESPVR consistency.

A

[0.1, 5.0] mmHg

EDPVR scaling
factor. Sets
baseline pressure.

EDP

[1, 45] mmHg

End-diastolic
pressure. Primary
clinical output.

Diastolic Physiology Model

e' (annular velocity)

$$e' = 13 * \sqrt{\tau_{ref} / \tau} \\ * \exp(-8 * (E_{ed} - 0.05))$$

Relaxation (τ) + Stiffness (E_{ed})

E (transmitral velocity)

$$\Delta P = LAP - P_m \text{ in} \\ E = 18 * \sqrt{\Delta P} * \text{eff}(\tau)$$

Bernoulli + pseudonormalization

$$E/e' = E / e'$$

Emergent, not prescribed

Same $E/e' = 10$ maps to different EDP depending on τ , E_{ed} , EDV. PINN disambiguates using all 11 inputs.

Systolic & Arterial Elastance

Ees — Systolic Elastance

Chen/Shishido approximation:

$$\text{Chen_Ees} = \text{ESP} / |\text{ESV} - 0.1 * \text{EDV}|$$

$$\text{Ees} = \text{Chen_Ees} * (1 + 0.3 * \tanh(d))$$

Physics-anchored estimate with a learned correction factor (delta) from the encoder.

Range: [0.3, 10.0] mmHg/ml

Ea — Arterial Elastance

Gated blend mechanism:

$$\text{Ea_phys} = \text{ESP} / \text{SV}$$

$$\text{Ea_learn} = \text{sigmoid}(\text{raw}) * [0.3, 6.0]$$

$$\text{gate} = \text{sigmoid}(\text{latent}[6])$$

$$\begin{aligned} \text{Ea} = & \text{gate} * \text{Ea_learn} \\ & + (1 - \text{gate}) * \text{Ea_phys} \end{aligned}$$

Learns when to trust physics vs. data.

Training Pipeline

1

Simulator

6 cardiac phenotypes
(Normal, HFrEF, HFpEF,
HTN, Elderly, Athlete)
15,000 samples
with measurement noise

>
>

2

Standardize

StandardScaler on
11 input features.
Scaler saved in
checkpoint for
MIMIC inference.

>
>

3

Train

MSE on 7 targets
Weighted loss:
[1, 5, 1, 0.5, 3, 0.5, 2]
Eed x5, EDP x3, tau x2
SiLU + BN + Dropout

>
>

4

Validate

MIMIC-IV-ECHO
35,540 echo studies
PCWP (n=1,692)
DD grade as external
validation targets

Loss weights emphasize diastolic: Eed x5, EDP x3, tau x2 | Systolic/coupling at x1 / x0.5

Circularity Defense

1. EDPVR-Only EDP

$EDP = A * \exp(E_{ed} * (EDV - V_0))$. Zero contribution from E/e' in the equation.
No Nagueh formula anywhere in decoder.

2. Emergent E/e'

Simulator generates E/e' from τ and E_{ed} via diastolic physiology. Not from inverse Nagueh $(EDP - 1.9) / 1.24$.

3. E/e' Dropout

30% masking of E/e' during training. Forces model to rely on EDV, ESV, HR, CO for EDP estimation when E/e' is absent.

MIMIC-IV Validation Results

0.688

Eed AUC
PCWP ≥ 18

0.650

DD Grade
Spearman rho

+0.009

Incremental
over E/e' alone

35,540

Echo Studies
MIMIC-IV

Model	Eed AUC	EDP AUC	DD rho
E/e' alone	0.738	--	--
PINN v41 (best)	0.688	0.655	0.650
E/e' + PINN Eed	0.746	--	--

Summary & Next Steps

- Physics-embedded decoder: EDPVR equation replaces learned shortcuts
- Emergent E/e' from diastolic physiology (τ , E_{ed}) breaks Nagueh tautology
- Novel non-invasive τ estimation — relaxation time constant as output
- Gated E_a : learns when to trust physics (ESP/SV) vs. encoder correction
- MIMIC-IV validation: 35,540 studies, E_{ed} AUC 0.688, DD ρ 0.650

Next Steps

1. IEEE JBHI submission — methods novelty focus (single MIMIC cohort)
2. MESA external validation via BioLINCC (IRB + application in parallel)
3. npj Digital Medicine — with MESA multi-ethnic external validation